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絶対値のついた関数の定積分

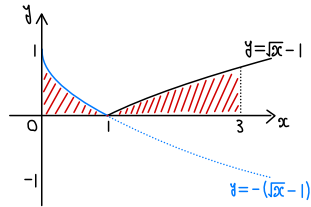
(例1) 次の定積分を求めよ。

(1) $\int_0^3 |\sqrt{x}-1| dx$

$$|\sqrt{x}-1| = \begin{cases} -(\sqrt{x}-1) & (0 \leq x \leq 1) \\ \sqrt{x}-1 & (1 \leq x \leq 3) \end{cases}$$

であるから

$$\begin{aligned} \int_0^3 |\sqrt{x}-1| dx &= \int_0^1 \{-(\sqrt{x}-1)\} dx + \int_1^3 (\sqrt{x}-1) dx \\ &= -\left[\frac{2}{3}x^{\frac{3}{2}}-x\right]_0^1 + \left[\frac{2}{3}x^{\frac{3}{2}}-x\right]_1^3 \\ &= -\left(\frac{2}{3}-1\right) + \left\{(2\sqrt{3}-3)-\left(\frac{2}{3}-1\right)\right\} \\ &= 2\sqrt{3}-\frac{7}{3} \end{aligned}$$

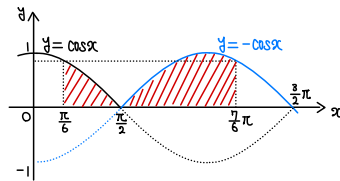


(2) $\int_{\frac{\pi}{6}}^{\frac{7}{6}\pi} |\cos x| dx$

$$|\cos x| = \begin{cases} \cos x & (\frac{\pi}{6} \leq x \leq \frac{\pi}{2}) \\ -\cos x & (\frac{\pi}{2} \leq x \leq \frac{7}{6}\pi) \end{cases}$$

であるから

$$\begin{aligned} \int_{\frac{\pi}{6}}^{\frac{7}{6}\pi} |\cos x| dx &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cos x dx + \int_{\frac{\pi}{2}}^{\frac{7}{6}\pi} (-\cos x) dx \\ &= \left[\sin x\right]_{\frac{\pi}{6}}^{\frac{\pi}{2}} - \left[\sin x\right]_{\frac{\pi}{2}}^{\frac{7}{6}\pi} \\ &= \left(1-\frac{1}{2}\right) - \left(-\frac{1}{2}-1\right) \\ &= 2 \end{aligned}$$



(例2) 定積分 $I = \int_0^{\pi} |\sin x + \cos x| dx$ を求めよ。

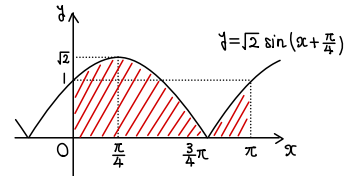
$$\begin{aligned} |\sin x + \cos x| &= |\sqrt{2} \sin(x + \frac{\pi}{4})| \\ &= \begin{cases} \sqrt{2} \sin(x + \frac{\pi}{4}) & (0 \leq x \leq \frac{3}{4}\pi) \\ -\sqrt{2} \sin(x + \frac{\pi}{4}) & (\frac{3}{4}\pi \leq x \leq \pi) \end{cases} \end{aligned}$$

であるから

$$\begin{aligned} I &= \int_0^{\frac{3}{4}\pi} \sqrt{2} \sin(x + \frac{\pi}{4}) dx + \int_{\frac{3}{4}\pi}^{\pi} \{-\sqrt{2} \sin(x + \frac{\pi}{4})\} dx \\ &= \sqrt{2} \left\{ \int_0^{\frac{3}{4}\pi} \sin(x + \frac{\pi}{4}) dx - \int_{\frac{3}{4}\pi}^{\pi} \sin(x + \frac{\pi}{4}) dx \right\} \end{aligned}$$

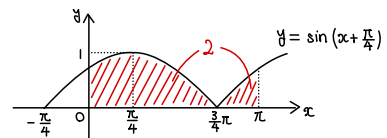
ここで、 $F(x) = -\cos(x + \frac{\pi}{4})$ とおくと

$$\begin{aligned} I &= \sqrt{2} \left\{ F(\frac{3}{4}\pi) - F(0) \right\} - \left\{ F(\pi) - F(\frac{3}{4}\pi) \right\} \\ &= \sqrt{2} \left\{ 2F(\frac{3}{4}\pi) - F(0) - F(\pi) \right\} \\ &= \sqrt{2} \left\{ 2 \cdot 1 - \left(-\frac{1}{\sqrt{2}}\right) - \frac{1}{\sqrt{2}} \right\} \\ &= 2\sqrt{2} \end{aligned}$$



(補足)

$$\begin{aligned} I &= \sqrt{2} \int_0^{\pi} |\sin(x + \frac{\pi}{4})| dx \\ &= 2\sqrt{2} \end{aligned}$$



(3) $\int_0^1 |e^x-2| dx$ ← $e^x-2=0$ より $x = \log 2$

$$|e^x-2| = \begin{cases} -(e^x-2) & (0 \leq x \leq \log 2) \\ e^x-2 & (\log 2 \leq x \leq 1) \end{cases}$$

であるから

$$\begin{aligned} \int_0^1 |e^x-2| dx &= \int_0^{\log 2} \{-(e^x-2)\} dx + \int_{\log 2}^1 (e^x-2) dx \\ &= -[e^x-2x]_0^{\log 2} + [e^x-2x]_{\log 2}^1 \\ &= -\{(2-2\log 2)-1\} + \{(e-2)-(2-2\log 2)\} \\ &= e+4\log 2-5 \end{aligned}$$

